

Mock Midterm — Form B

Time allowed: 120 minutes · 100 points

Name: _____ Section: _____ Date: _____

	Part I	Part II	Part III	Part IV	Part V	Total
Points	30	20	20	10	20	100
Score						

Instructions. Coverage: Modules 1–4. No calculator. Answer everything. Part III: draw on the grids and axes. Part IV: show complete solutions; final answers need units. Part V: draw, explain, or both. Unless a problem says otherwise, use the slide conventions (NRZ-L: 0 = +V; NRZ-I: a 1 inverts; Manchester: 0 = high→low; differential Manchester: 0 = transition at the start; AMI: first 1 is +V).

Part I — Multiple choice

30 questions × 1 pt = 30 pts

Write the letter of the best answer on the line before each number.

- _____ 1. The receiver's clock runs **faster** than the sender's clock. What happens?
- A. Nothing — the receiver re-times itself automatically
 - B. The receiver misses bits and sees a shorter stream
 - C. The bits arrive in a different order
 - D. The receiver reads extra bits (for example 1001 bps instead of 1000 bps) and misinterprets the stream
- _____ 2. On a constellation diagram, the horizontal (x) axis represents:
- A. time
 - B. the quadrature carrier
 - C. frequency
 - D. the in-phase carrier
- _____ 3. According to the slides, what does block coding do for the line code that follows it?
- A. It lowers the bit rate by 20%
 - B. It removes the need for a line code
 - C. It makes every bit have a mid-bit transition
 - D. It adds redundancy so that bit combinations giving DC components or poor synchronization are avoided
- _____ 4. For a range of -10 V to $+10\text{ V}$ and $L = 4$ zones, what is the zone height Δ ?
- A. 2.5 V
 - B. 5 V
 - C. 10 V
 - D. 20 V
- _____ 5. A complex bandpass signal has a bandwidth of 300 kHz. What is its minimum sampling rate?
- A. 600,000 samples/s
 - B. 300,000 samples/s
 - C. It cannot be determined from the bandwidth alone
 - D. 150,000 samples/s

- _____ 6. What does **error detection** mean in the context of line coding?
- A. The receiver corrects the corrupted bit automatically
 - B. A signal pattern that is not part of the code arrives, so the receiver knows a symbol error occurred
 - C. The signal is immune to noise and cannot be corrupted
 - D. The sender repeats every bit three times
- _____ 7. A band from 600 kHz to 700 kHz is shared for full-duplex ASK. Where are the two carriers?
- A. 625 kHz and 675 kHz
 - B. 600 kHz and 700 kHz
 - C. 650 kHz for both
 - D. 612.5 kHz and 687.5 kHz
- _____ 8. A 16-QAM signal carries 20 Mbps with $d = 0$. What bandwidth does it need?
- A. 20 MHz
 - B. 10 MHz
 - C. 5 MHz
 - D. 80 MHz
- _____ 9. What is the propagation time over a 3,000 km link if signals travel at 2×10^8 m/s?
- A. 1.5 ms
 - B. 6 ms
 - C. 15 ms
 - D. 150 ms
- _____ 10. According to Shannon, the capacity of a channel depends on:
- A. the bandwidth and the number of signal levels
 - B. the number of signal levels and the SNR
 - C. the propagation speed and the distance
 - D. the bandwidth and the signal-to-noise ratio
- _____ 11. A noisy channel has a bandwidth of 2 MHz and an SNR of 15. What is its Shannon capacity?
- A. 8 Mbps
 - B. 30 Mbps
 - C. 4 Mbps
 - D. 2 Mbps
- _____ 12. In **MLT-3**, the signal is at +V and the next bit is a 1. Where does the level go?
- A. -V
 - B. 0 V
 - C. It stays at +V
 - D. +2V
- _____ 13. A digital signal has 16 levels. How many bits does each level carry?
- A. 16 bits
 - B. 8 bits
 - C. 4 bits
 - D. 2 bits
- _____ 14. The two styles of media access control named on the slides are:
- A. ALOHA style and token style
 - B. TCP and UDP
 - C. Analog and digital
 - D. Simplex and duplex
- _____ 15. Which pattern does **B8ZS** substitute for eight consecutive zeros?
- A. 000VB0VB
 - B. B00V
 - C. 000V
 - D. VB0VB000
- _____ 16. Which component of a PCM decoder smooths the held staircase back into an analog signal?
- A. A comparator
 - B. A delay unit
 - C. A quantizer
 - D. A low-pass filter

_____ 17. Why is a **DC component** a problem for line coding?

- A. It makes the baud rate higher than the bit rate
- B. It needs an extra wire to carry the clock
- C. Most channels are bandpass and may not support the low frequencies that a constant level creates
- D. It makes every bit have a mid-bit transition

_____ 18. Increasing the number of quantization levels L ...

- A. raises the quantization error
- B. lowers the quantization error but raises the bit rate
- C. lowers both the error and the bit rate
- D. has no effect on the bit rate

_____ 19. In the 4B/5B table, which 5-bit word replaces the data word **0000**?

- A. 01001
- B. 11101
- C. 11110
- D. 10101

_____ 20. A signal of 6 MHz travels in free space ($c = 3 \times 10^8$ m/s). What is its wavelength?

- A. 50 m
- B. 5 m
- C. 0.02 m
- D. 18 m

_____ 21. During a phone call you faintly hear a conversation from a neighbouring wire. This is:

- A. thermal noise
- B. crosstalk
- C. impulse noise
- D. induced noise

_____ 22. To carry 5 bits per signal element, how many distinct signal elements are needed?

- A. 5
- B. 10
- C. 25
- D. 32

_____ 23. Why can a composite signal be **distorted** on its way through a medium?

- A. The noise power is larger than the signal power
- B. An amplifier adds the same gain to every frequency
- C. It loses some of its energy in overcoming the resistance of the medium
- D. Its frequency components travel at different propagation speeds and arrive at different times

_____ 24. Which of these is a WAN technology mentioned on the slides?

- A. Bluetooth
- B. Submarine fiber cable
- C. A hub
- D. A coaxial patch cord

_____ 25. A data stream of rate N is sent with **2B1Q**. What is the average signal (baud) rate?

- A. N/4
- B. N/2
- C. N
- D. 2N

_____ 26. A link has a bandwidth of 10 Mbps. What can be said about the throughput through it?

- A. It is always equal to 10 Mbps
- B. It is always less than 10 Mbps
- C. It can be greater than 10 Mbps
- D. It is unrelated to the bandwidth

_____ 27. In the slides' example of digital transmission, a binary 1 can be represented as:

- A. -5 V
- B. 0 V
- C. +5 V
- D. a 5 kHz tone

_____ 28. Which of these is NOT a service the data link layer provides to the upper layers?

- | | |
|--|---------------------------|
| A. Framing | B. Flow control |
| C. Routing packets between networks | D. Error detection |

_____ 29. SpaceX Starlink satellites are deployed at about what altitude?

- | | |
|--------------------|---------------------|
| A. 55 km | B. 550 km |
| C. 5,500 km | D. 35,786 km |

_____ 30. In binary FSK, bit 1 is sent at which frequency?

- | | |
|----------------------------|------------------|
| A. $f_c - \Delta f$ | B. f_c |
| C. $f_c + \Delta f$ | D. $2f_c$ |

Part II — Identification

20 questions × 1 pt = 20 pts

Write the term on the answer line.

1. The device that converts digital signals into analog signals for transmission (its partner converts them back).

Answer: _____

2. The difference in delay (latency) from one packet to another.

Answer: _____

3. The polar NRZ version in which the *change or lack of change* in polarity gives the bit: a 1 inverts the level, a 0 does not.

Answer: _____

4. The data link layer service that ensures sufficient bandwidth is available and that delays are predictable and guaranteed.

Answer: _____

5. The main frequency of a composite signal (in the slide example the other sine waves have 3 and 9 times this frequency).

Answer: _____

6. The data link layer service that keeps frames in the correct order as they are exchanged.

Answer: _____

7. In the B8ZS pattern 000VB0VB, what does the letter **V** stand for?

Answer: _____

8. The type of network that uses unguided media and can be short or long distance.

Answer: _____

9. Serial transmission in which uneven gaps between frames are not allowed; bits flow with fixed, equal gaps.

Answer: _____

10. Sending a digital signal over a channel without changing it into an analog signal.

Answer: _____

- 11.** The digital-to-analog scheme that varies only the phase of the carrier and is much more robust than ASK.
Answer: _____
- 12.** The property of a line code that lets the receiver keep its clock in step with the sender's clock by using the transitions in the signal itself.
Answer: _____
- 13.** The drift of the receiver's running average (the baseline) caused by a long string of 0s or 1s, which makes correct decoding difficult.
Answer: _____
- 14.** The unit engineers use to show that a signal has lost or gained strength.
Answer: _____
- 15.** The device in an FSK transmitter whose output frequency follows its input voltage.
Answer: _____
- 16.** The name of the sampled signal — pulses whose heights are the (non-integer) sample values.
Answer: _____
- 17.** The PCM decoder circuit that holds each pulse amplitude until the next pulse arrives.
Answer: _____
- 18.** The French mathematician who showed that a composite signal is a combination of simple sine waves with different frequencies, amplitudes and phases.
Answer: _____
- 19.** The line-coding family in which *all* signal levels are on one side of the time axis (NRZ is the example).
Answer: _____
- 20.** The kind of FSK that switches frequency without adhering to the current phase of the signal.
Answer: _____

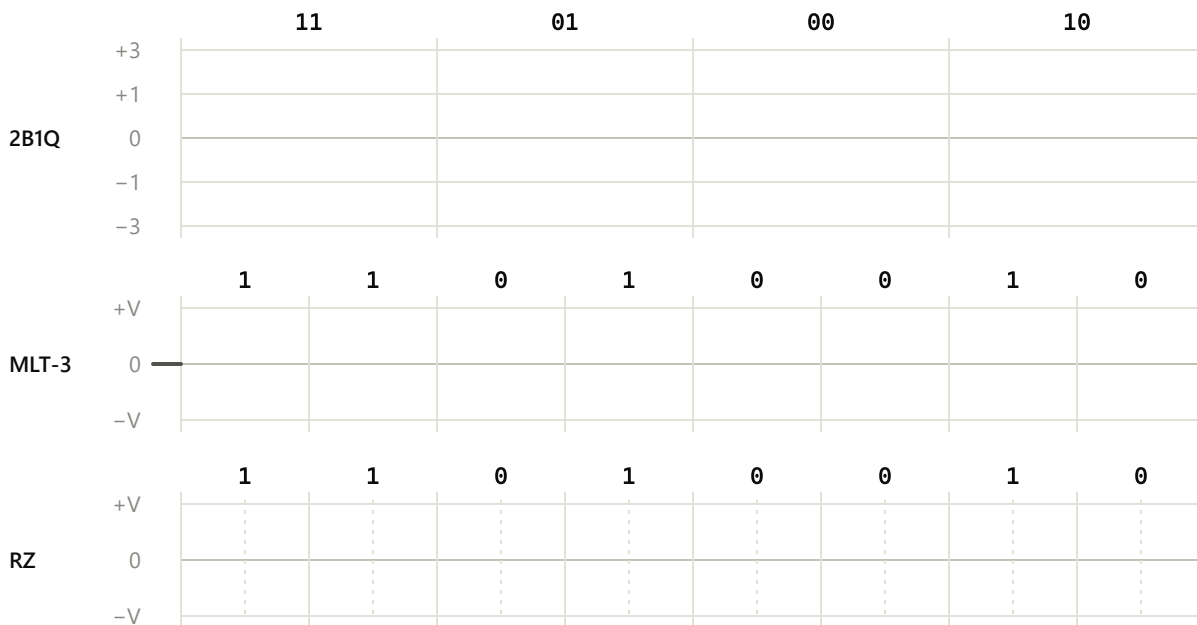
Part III — Draw: line encodings and waves

4 problems × 5 pts = 20 pts

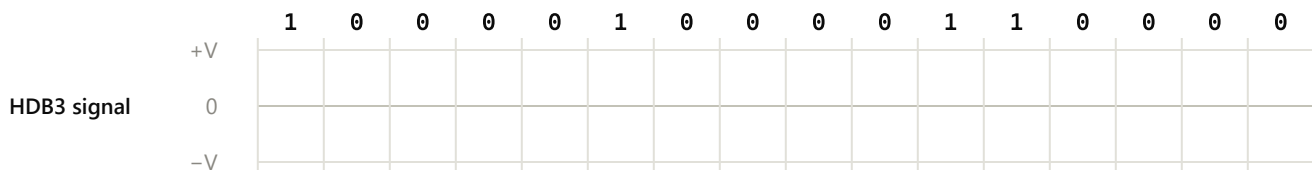
Show a complete solution; give final answers with units.

1. Draw the bit stream **11010010** as a waveform in each of the line codes below (fill every cell of the grid). Use the slide conventions: [5 pts]

- 2B1Q: one level per pair of bits from the transition table (previous level positive: 00 → +1, 01 → +3, 10 → -1, 11 → -3; previous level negative: the signs reversed); assume the level before the first pair is positive.
- MLT-3: a 0 keeps the level; a 1 moves round the cycle 0 → +V → 0 → -V → ...; start at 0 V with the last nonzero level -V, so the first 1 goes to +V.
- RZ: a 1 is +V then 0 V, a 0 is -V then 0 V (every bit returns to zero in the middle).



2. Apply **HDB3** scrambling (on top of AMI) to the data stream **1000 0100 0011 0000** and draw the line signal. [5 pts]
The last nonzero pulse sent before this stream was positive (+V). At the start of the stream the number of nonzero pulses since the last substitution is even.



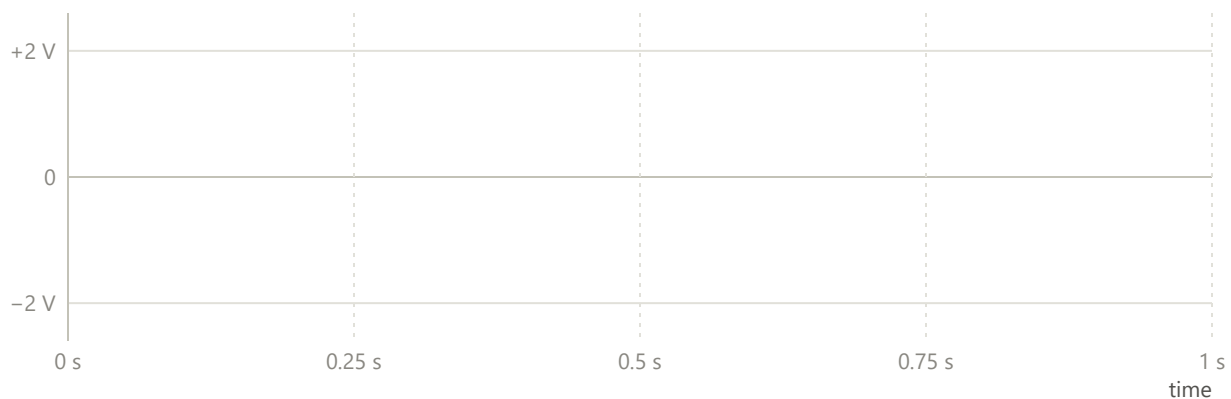
3. A signal between -8 V and 8 V is quantized into $L = 4$ zones (zone height $\Delta = 4\text{ V}$). Its samples at $t_1 \dots t_6$ are **5.4, -1.2, -6.6, 2.9, 7.1, -3.3 V**. Draw the quantized signal: each sample moved to its zone midpoint and held until the next sample. [5 pts]

Quantized signal



4. Draw one second of a sine wave with peak amplitude 2 V , frequency 3 Hz and phase 270° . [5 pts]

Sine wave, 0 to 1 s



Part IV — Problem solving (no calculator)

2 problems × 5 pts = 10 pts

Show a complete solution; give final answers with units.

1. We have a channel with a **4,000 Hz** bandwidth. The SNR for this channel is **63**. Find the upper limit of the bit rate. For better performance we then use a lower bit rate of **16,000 bps**: how many signal levels are needed? [5 pts]

WORK

Upper limit $C =$ _____ bps

Signal levels $L =$ _____

2. We have an available bandwidth of **120 kHz which spans from 500 kHz to 620 kHz**. What should be the carrier frequency and the bit rate if we modulate our data with **FSK**, $d = 0.5$, and $2\Delta f = 30$ kHz? Also give the frequencies used for bit 1 and bit 0. [5 pts]

WORK

Carrier $f_c =$ _____ Hz

Bit rate $N =$ _____ bps

f_1 (bit 1) = _____ Hz

f_2 (bit 0) = _____ Hz

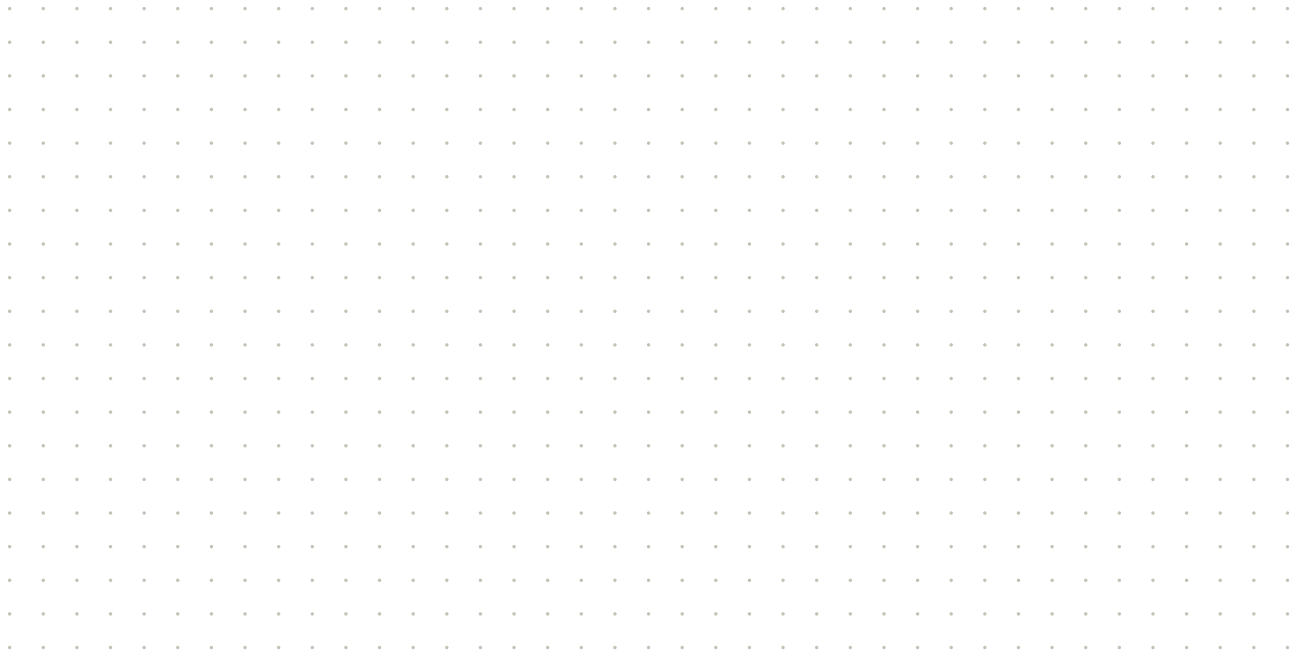
Part V — Essay (draw and/or explain)

4 questions × 5 pts = 20 pts

Answer in the space provided — draw, explain, or both.

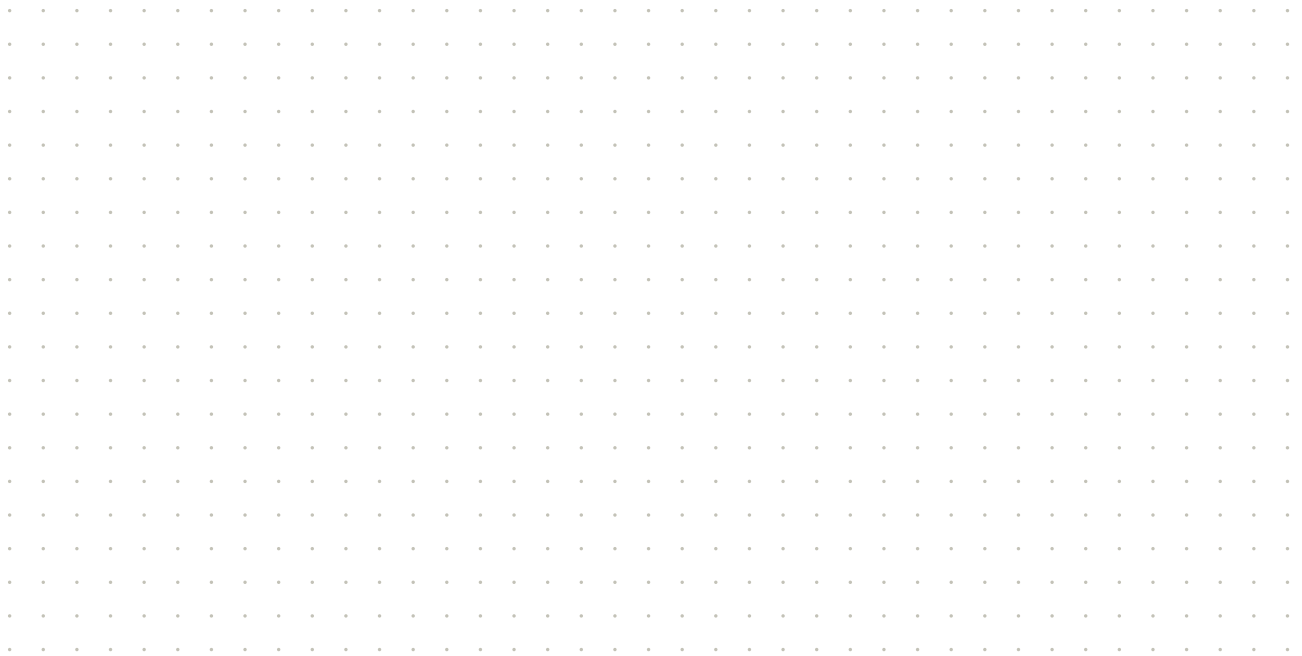
1. Explain the purpose of the data link layer and describe at least four services it provides to the upper layers. [5 pts]
Then explain why digital data is sometimes sent as an analog signal.

DRAW AND/OR EXPLAIN



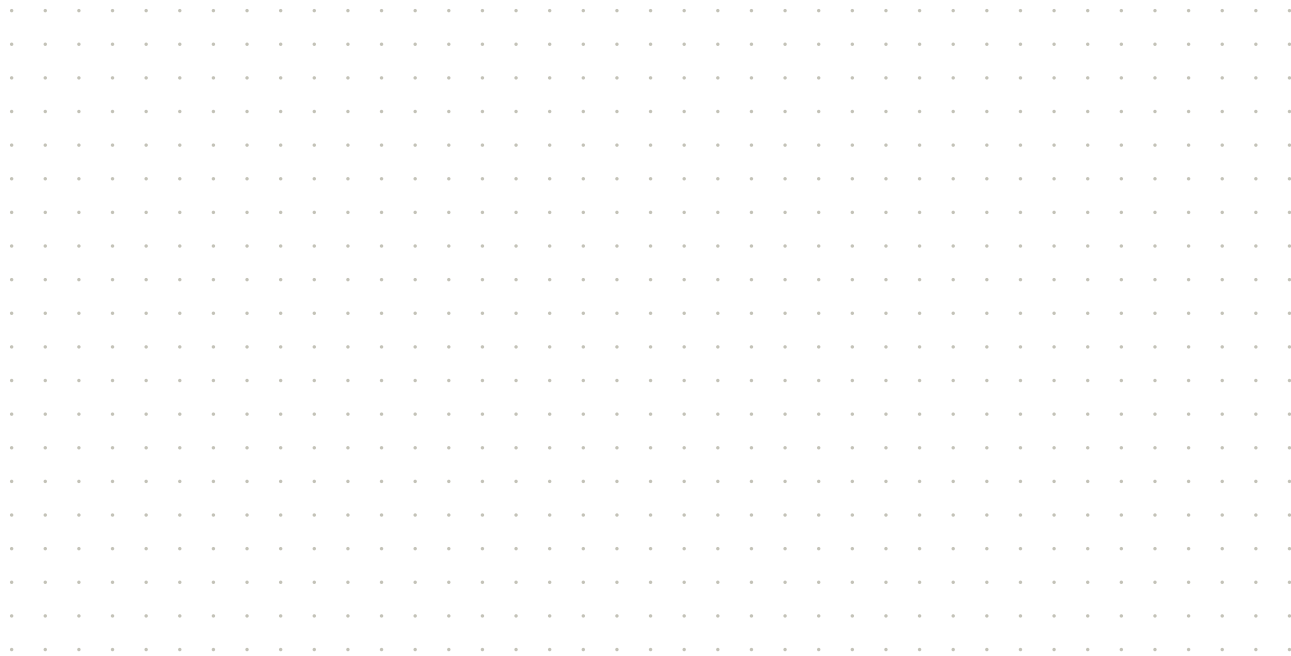
2. Compare ASK, FSK and PSK: what each changes in the carrier and how each one's bandwidth depends on the baud rate; then say how robust ASK and PSK are to noise. Then explain how QPSK sends twice the bits of BPSK without needing more bandwidth. Sketches of the three signals for a short bit string are welcome. [5 pts]

DRAW AND/OR EXPLAIN



3. Why does **AMI** need scrambling? Explain how **B8ZS** and **HDB3** work, and draw B8ZS for the bits **100000000** [5 pts]
when the last pulse sent before this stream was $-V$ (so the 1 is sent as $+V$). Mark each V and B.

DRAW AND/OR EXPLAIN



4. Describe the three transmission impairments named in the lecture: **attenuation**, **distortion** and **noise**. For [5 pts]
each, draw (or describe) what it does to a signal and say what causes it; then explain how losses, gains and noise are
measured.

DRAW AND/OR EXPLAIN

